MATH 345 Skeleton Notes
Unit 2: Anatomy of a linear map
Section 2.5: Inverses and determinants

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Matrix invertibility and linear systems

In this section we finish the square-matrix version of the Unit 2 story. A square matrix is invertible exactly when it "does not forget any nonzero input direction" and "every output is reached once". The inverse matrix gives the inverse map when it exists. The determinant, a fundamental quantity in linear algebra that we only briefly discuss in this course, gives a single-number test for invertibility.

Definition: Invertibility of Matrices

An $n \times n$ matrix A is called nonsingular, or invertible, if there exists an $n \times n$ matrix B such that $AB = BA = I_n$. Such a B is called the inverse of A. If no such B exists, A is called singular or noninvertible.

Note

The inverse of a matrix mimics the reciprocal of a real number.

Activity: Testing whether two matrices are inverses (U2-LO5)

Let

$$A = \begin{bmatrix} 1 & 3 \\ 1 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 4 & -3 \\ -1 & 1 \end{bmatrix}.$$

Compute both AB and BA, and make a conclusion using the language of inverses.

Fact

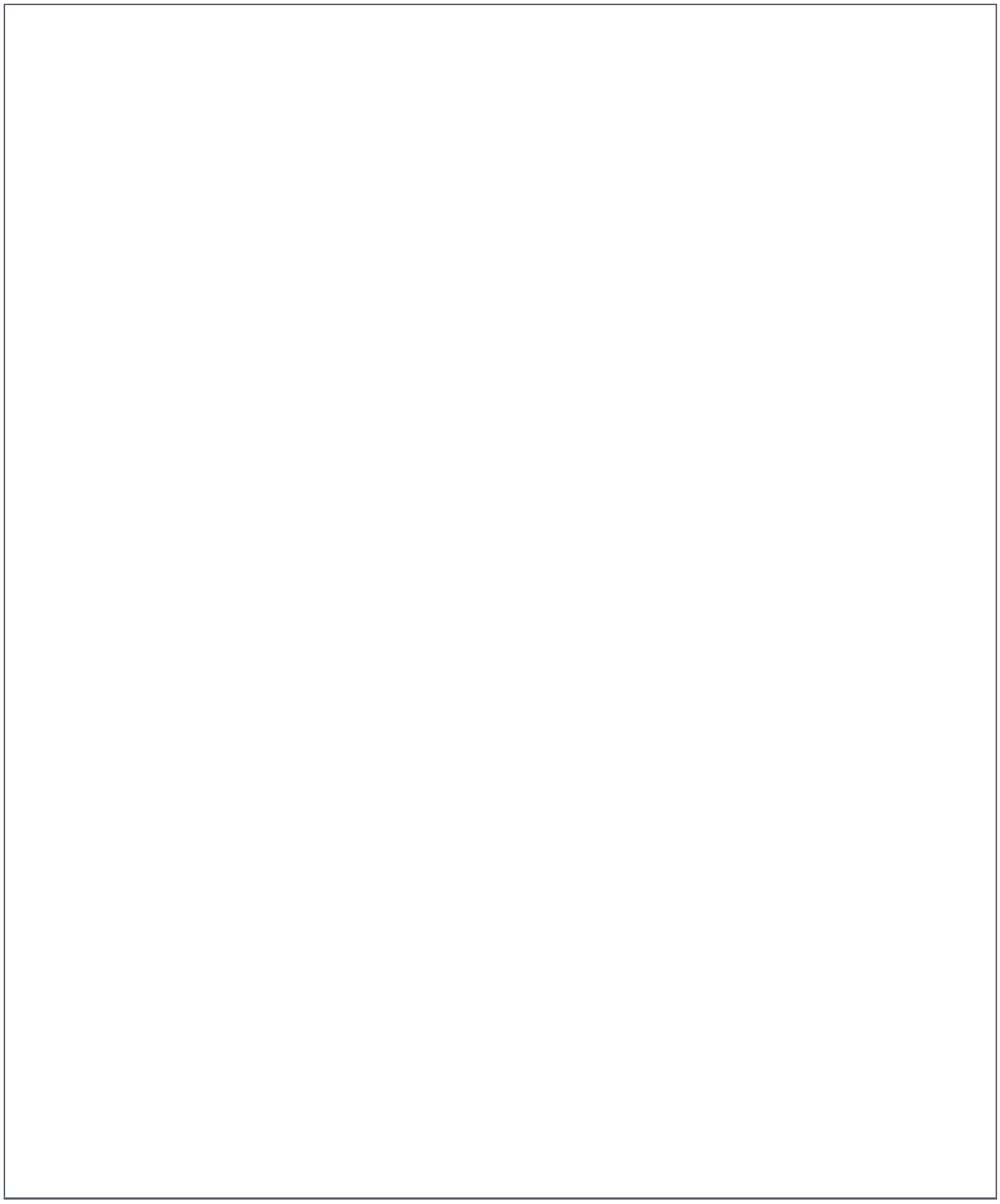
The inverse of a matrix, if it exists, is unique. Therefore, we can write A^{-1} for the inverse of A.

Fact

For $n \times n$ matrices A and B, if $AB = I_n$, then $BA = I_n$.

Activity: Testing a matrix for invertibility (U2-LO5)

Does $A = \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}$ have an inverse?



Theorem

If both A and B are nonsingular $n \times n$ matrices, then the matrix AB is

nonsingular and its inverse is $(AB)^{-1} = B^{-1}A^{-1}$.

Why is this true?
It suffices to calculate that $(AB)(B^{-1}A^{-1}) = I_n$ and $(B^{-1}A^{-1})(AB) = I_n$. Firstly,

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AI_nA^{-1} = AA^{-1} = I_n.$$

Similarly,

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}I_nB = B^{-1}B = I_n.$$

Fact: Follow Up Facts About Invertibility

- If $A_1, A_2, \dots, A_k$ are $n \times n$ invertible / nonsingular matrices, then $A_1A_2 \cdots A_k$ is invertible / nonsingular and

$$(A_1A_2 \cdots A_k)^{-1} = A_k^{-1} \cdots A_2^{-1}A_1^{-1}.$$

- If A is invertible/nonsingular, then A^{-1} is invertible/nonsingular and

$$(A^{-1})^{-1} = A.$$

- If A is invertible/nonsingular, then A^T is invertible/nonsingular and

$$(A^{-1})^T = (A^T)^{-1}.$$

Let $T = T_A : \mathbb{R}^n \to \mathbb{R}^n$ be a matrix transformation induced by the $n \times n$ matrix A. What does invertibility imply about the geometry of the transformation T_A?
Consider $T' = T_{A^{-1}}$. What happens when we compose them?

Activity: Composing a map and its inverse (U2-LO5)

Compute the composition of T and T' in both orders.

Definition

Given a matrix transformation $T : \mathbb{R}^n \to \mathbb{R}^n$, if there exists a transformation $S : \mathbb{R}^n \to \mathbb{R}^n$ such that $S(T(\mathbf{x})) = T(S(\mathbf{x})) = \mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$, we say that T has an inverse.

Theorem

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ denote the matrix transformation induced by an $n \times n$ matrix A. Then:

1. A is invertible if and only if T has an inverse.
2. In this case, T has exactly one inverse (which we denote as T^{-1}), and $T^{-1} : \mathbb{R}^n \to \mathbb{R}^n$ is the transformation induced by the matrix A^{-1}.

In other words, $(T_A)^{-1} = T_{A^{-1}}$.
T and T^{-1} 'reverse' or 'undo' the action of each other.

Activity: Reversing a reflection (U2-LO5)

Consider $T : \mathbb{R}^2 \to \mathbb{R}^2$ given by reflection across the x-axis. Find the matrix representing this transformation, and find the inverse.

Theorem: Determinant and inverse of a 2×2 matrix

The 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

is invertible if and only if $ad - bc \neq 0$, in which case

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

We define the **determinant** of the matrix A to be the quantity $\det(A) = ad - bc$. Thus the matrix A is invertible if and only if its determinant is nonzero.

For now, $\det(A) = ad - bc$ is only the determinant of a 2×2 matrix. We will define determinants for larger square matrices later in this section.

Activity: Reading determinant checks in code (U2-LO3, U2-LO5, U2-LO7)

The following code computes determinants of two 2×2 matrices.

```
import numpy as np

A = np.array([
    [1., 1.],
    [0., 1.],
])

B = np.array([
    [1., 0.],
    [0., 0.],
])

detA = float(np.linalg.det(A))
detB = float(np.linalg.det(B))

detA, detB
```

Output:

```
(1.0, 0.0)
```

Activity: Reading determinant checks in code (U2-LO3, U2-LO5, U2-LO7) (continued)

1. Which matrix is invertible?
2. Which matrix forgets a nonzero input direction?
3. How does the determinant tell you?

Activity: Checking invertibility and finding inverses (U2-LO5)

Find the inverse of each matrix, if possible.

$$A = \begin{bmatrix} -2 & -3 \\ 4 & 6 \end{bmatrix}$$

Activity: Checking invertibility and finding inverses (U2-LO5)

Find the inverse of each matrix, if possible.

$$B = \begin{bmatrix} 1 & 6 \\ 2 & 3 \end{bmatrix}$$

If A is an $n \times n$ matrix, then the linear system $A\mathbf{x} = \mathbf{b}$ is a system of n equations in n unknowns. Suppose A is nonsingular. How can we use A^{-1} to solve the system $A\mathbf{x} = \mathbf{b}$?

Activity: Solving a system with an inverse (U2-LO5)

Solve the system using the inverse of A.

Consequences:

- When A^{-1} exists, then $A\mathbf{x} = \mathbf{b}$ has a unique solution.
- If A is invertible/nonsingular, then the *only* solution to the homogeneous system $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$.
- Put differently, an invertible square matrix forgets no nonzero input direction.

Activity: One inverse, several right-hand sides (U2-LO5)

Suppose that

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Use this information to calculate the solution to the following systems:
$A\mathbf{x} = \mathbf{b}$

Activity: One inverse, several right-hand sides (U2-LO5)

Suppose that

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Use this information to calculate the solution to the following systems:

$A\mathbf{x} = \mathbf{c}$

Activity: One inverse, several right-hand sides (U2-LO5)

Suppose that

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Use this information to calculate the solution to the following systems:

$A\mathbf{x} = \mathbf{0}$

Activity: A coordinate map and invertibility (U2-LO5)

Consider the transformation $P : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$P\left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}\right) = \begin{bmatrix} a_1 \\ 0 \end{bmatrix}.$$

Is P a matrix transformation? If so, find the matrix A corresponding to P.

Activity: A coordinate map and invertibility (U2-LO5)

Consider the transformation $P : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

Activity: A coordinate map and invertibility (U2-LO5) (continued)

$$P\left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}\right) = \begin{bmatrix} a_1 \\ 0 \end{bmatrix}.$$

Is P an invertible transformation?

Consider vectors $\mathbf{c}_1, \ldots, \mathbf{c}_n$ in $\mathbb{R}^m$, and let A be the $m \times n$ matrix whose columns are the vectors $\mathbf{c}_1, \ldots, \mathbf{c}_n$. Recall that $\{\mathbf{c}_1, \ldots, \mathbf{c}_n\}$ are linearly independent precisely when the equation $A\mathbf{x} = \mathbf{0}$ has *only* the *trivial* solution (see Linear Independence and the uniqueness theorem, Uniqueness of linear-combination coefficients). By Spanning and solvability, $\text{span}\{\mathbf{c}_1, \ldots, \mathbf{c}_n\} = \mathbb{R}^m$ exactly when $A\mathbf{x} = \mathbf{b}$ has a solution for all $\mathbf{b}$ in $\mathbb{R}^m$.

Theorem: Invertibility Criteria

Let A be an $n \times n$ matrix. The following are equivalent:

1. A is nonsingular/invertible.
2. The columns of A are linearly independent.
3. The columns of A span $\mathbb{R}^n$.
4. The rows of A are linearly independent.
5. The rows of A span $\mathbb{R}^n$.
6. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
7. A is row equivalent to I_n.

8. The linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $n \times 1$ matrix $\mathbf{b}$.

9. The rank of A is n.

Activity: Testing dependence using a matrix (U2-LO4)

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 3 \\ 0 \\ -5 \end{bmatrix}.$$

Is S linearly dependent?

General Determinants

We generalize the determinant to $n \times n$ matrices, which in particular gives us a way to determine whether a matrix is invertible. We use the 2×2 determinant from Determinant and inverse of a 2×2 matrix to define the 3×3 determinant. Later, an $n \times n$ determinant will be built from $(n-1) \times (n-1)$ determinants.

Definition: Minors of a 3×3 matrix

Let A be a 3×3 matrix. The (i, j)-th minor of A, denoted

$$\text{minor}(A)_{ij},$$

is the determinant of the 2×2 matrix obtained from A by deleting row i and column j.

Activity: Computing matrix minors (U2-LO7)

Suppose

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}.$$

Calculate $\text{minor}(A)_{11}$.

Activity: Computing matrix minors (U2-LO7)

Suppose

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}.$$

Calculate $\text{minor}(A)_{23}$.

Definition: Cofactors of a 3×3 matrix

Let A be a 3×3 matrix. The (i, j)-cofactor of A, denoted

$$\text{cof}(A)_{ij},$$

is

$$\text{cof}(A)_{ij} = (-1)^{i+j} \, \text{minor}(A)_{ij}.$$

Activity: Computing cofactors (U2-LO7)

Define

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}$$

Compute $\text{cof}(A)_{11}$.

Activity: Computing cofactors (U2-LO7)

Define

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}$$

Compute $\operatorname{cof}(A)_{23}$.

Definition: Determinant of a 3×3 matrix

Let $A = [a_{ij}]$ be a 3×3 matrix. The determinant of A, denoted $\det(A)$, is defined by expansion along the first row:

$$\det(A) = a_{11} \operatorname{cof}(A)_{11} + a_{12} \operatorname{cof}(A)_{12} + a_{13} \operatorname{cof}(A)_{13}.$$

This process is called cofactor expansion.

Fact

For a 3×3 matrix, cofactor expansion along any row or any column gives the same determinant.

For example, expanding along the second row means multiplying each entry in that row by its cofactor and adding:

$$\det(A) = a_{21}\,\mathrm{cof}(A)_{21} + a_{22}\,\mathrm{cof}(A)_{22} + a_{23}\,\mathrm{cof}(A)_{23}.$$

Expanding along the second column instead means multiplying each entry in that column by its cofactor and adding:

$$\det(A) = a_{12}\,\mathrm{cof}(A)_{12} + a_{22}\,\mathrm{cof}(A)_{22} + a_{32}\,\mathrm{cof}(A)_{32}.$$

Activity: Comparing row and column expansions (U2-LO7)

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}.$$

Find the determinant of A by expanding along row 1.

Activity: Comparing row and column expansions (U2-LO7)

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}.$$

Find the determinant of A by expanding along column 2.

Note

To track the sign $(-1)^{i+j}$ when computing cofactors, it may be helpful to view the matrix as a checkerboard as below:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

The sign in the ith row and jth position with be the sign of $(-1)^{i+j}$.

Note

Since any row or column gives the same determinant, choose a row or column with many zeros when possible.

Activity: Choosing a determinant expansion (U2-LO7)

Compute the determinant of the following matrices:

$$A = \begin{bmatrix} 3 & -5 & 1 \\ 0 & 1 & 0 \\ -2 & -4 & 7 \end{bmatrix}$$

Activity: Choosing a determinant expansion (U2-LO7)

Compute the determinant of the following matrices:

$$B = \begin{bmatrix} 0 & \pi & 200 \\ 0 & 0.03 & 4 \\ 0 & 19 & -3584 \end{bmatrix}$$

Note

Whenever a row or column of a matrix only has zeros as entries, expanding along that row shows that the determinant of the matrix is always zero, so no cofactor computations are required.

Now that 3×3 determinants have been defined, we can define larger determinants recursively. For a 4×4 matrix, each minor determinant is a 3×3 determinant. For a 5×5 matrix, each minor determinant is a 4×4 determinant, and so on.

Definition: Determinant of a general square matrix

Let $A = [a_{ij}]$ be an $n \times n$ matrix with $n \geq 4$. Assume determinants have already been defined for $(n-1) \times (n-1)$ matrices.

The (i, j)-th minor of A, denoted

$$\operatorname{minor}(A)_{ij},$$

is the determinant of the $(n-1) \times (n-1)$ matrix obtained from A by deleting row i and column j.

The (i, j)-cofactor of A is

$$\operatorname{cof}(A)_{ij} = (-1)^{i+j} \operatorname{minor}(A)_{ij}.$$

The determinant of A is defined by expansion along the first row:

$$\det(A) = \sum_{j=1}^{n} a_{1j} \operatorname{cof}(A)_{1j}.$$

Fact

For an $n \times n$ matrix, cofactor expansion along any row or any column gives the same determinant:

$$\det(A) = \sum_{j=1}^{n} a_{ij} \operatorname{cof}(A)_{ij}$$

for any fixed row i, and

$$\det(A) = \sum_{i=1}^{n} a_{ij} \operatorname{cof}(A)_{ij}$$

for any fixed column j.

Example: A four-by-four determinant (U2-LO7)

Calculate the determinant of the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 4 & 2 & 3 \\ 1 & 3 & 4 & 5 \\ 3 & 4 & 3 & 2 \end{bmatrix}$$

Expand along column 3:

$$\det(A) = 3\,\mathrm{cof}(A)_{13} + 2\,\mathrm{cof}(A)_{23} + 4\,\mathrm{cof}(A)_{33} + 3\,\mathrm{cof}(A)_{43}.$$

The signs in column 3 are

$$+,\ -,\ +,\ -.$$

So

$$\begin{aligned}\det(A) = 3\det\begin{bmatrix}5 & 4 & 3\\ 1 & 3 & 5\\ 3 & 4 & 2\end{bmatrix}\\ -2\det\begin{bmatrix}1 & 2 & 4\\ 1 & 3 & 5\\ 3 & 4 & 2\end{bmatrix}\\ +4\det\begin{bmatrix}1 & 2 & 4\\ 5 & 4 & 3\\ 3 & 4 & 2\end{bmatrix}\\ -3\det\begin{bmatrix}1 & 2 & 4\\ 5 & 4 & 3\\ 1 & 3 & 5\end{bmatrix}.\end{aligned}$$

Expanding each 3×3 determinant along its first row,

$$\begin{aligned}\det\begin{bmatrix}5 & 4 & 3\\ 1 & 3 & 5\\ 3 & 4 & 2\end{bmatrix} &= 5(3\cdot 2 - 5\cdot 4)\\ &\quad - 4(1\cdot 2 - 5\cdot 3) + 3(1\cdot 4 - 3\cdot 3)\\ &= -33,\end{aligned}$$

$$\det\begin{bmatrix}1 & 2 & 4\\ 1 & 3 & 5\\ 3 & 4 & 2\end{bmatrix} = 1(3\cdot 2 - 5\cdot 4) - 2(1\cdot 2 - 5\cdot 3) + 4(1\cdot 4 - 3\cdot 3) = -8,$$

$$\det\begin{bmatrix}1 & 2 & 4\\ 5 & 4 & 3\\ 3 & 4 & 2\end{bmatrix} = 1(4\cdot 2 - 3\cdot 4) - 2(5\cdot 2 - 3\cdot 3) + 4(5\cdot 4 - 4\cdot 3) = 26,$$

and

$$\det\begin{bmatrix}1 & 2 & 4\\ 5 & 4 & 3\\ 1 & 3 & 5\end{bmatrix} = 1(4\cdot 5 - 3\cdot 3) - 2(5\cdot 5 - 3\cdot 1) + 4(5\cdot 3 - 4\cdot 1) = 11.$$

Therefore

$$\begin{aligned}\det(A) &= 3(-33) - 2(-8) + 4(26) - 3(11)\\ &= -99 + 16 + 104 - 33\\ &= -12.\end{aligned}$$

Activity: The determinant of a triangular matrix (U2-LO7)

Compute $\det(A)$, where

$$A = \begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}.$$

Recall Main diagonal.

Definition: Triangular and Diagonal Matrices

An $n \times n$ matrix $A = [a_{ij}]$ is called upper triangular if $a_{ij} = 0$ for $i > j$, and lower triangular if $a_{ij} = 0$ for $i < j$. A matrix is diagonal if it is both upper triangular and lower triangular, i.e., if $a_{ij} = 0$ for $i \neq j$. We denote the $n \times n$ diagonal matrix with entries $\lambda_1, \ldots, \lambda_n$ on the diagonal by $\text{diag}(\lambda_1, \ldots, \lambda_n)$.

Theorem

If $A = [a_{ij}]$ is a upper or lower triangular matrix, then $\det(A) = a_{11} \cdots a_{nn}$. That is, the determinant of a triangular matrix is the product of entries in the *main diagonal*.

Why is this true?

A generalization of the argument used to calculate The determinant of a triangular matrix (U2-LO7), i.e., expanding the determinant in cofactors along the first column iteratively.

Theorem

If a row or column of a matrix A consists entirely of zeros, then $\det(A) = 0$.

Why is this true?

A generalization of the argument used to calculate the determinant in Activity the activity.

Theorem: Determinant of a product

If A and B are $n \times n$ matrices, then $\det(AB) = \det(A)\det(B)$.

Activity: Determinants of matrices and their products (U2-LO7)

Let

$$A = \begin{bmatrix} 3 & 1 \\ -2 & 7 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 4 \\ 0 & 2 \end{bmatrix}.$$

Compute the determinant of A, B, and C.

Activity: Determinants of matrices and their products (U2-LO7)

Let

$$A = \begin{bmatrix} 3 & 1 \\ -2 & 7 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 4 \\ 0 & 2 \end{bmatrix}.$$

Compute the determinant of AC and BC.

Theorem

If A is a $n \times n$ matrix, then A is *nonsingular* if and only if $\det(A) \neq 0$. In this case $\det(A^{-1}) = 1/\det(A)$.

Why is this true?

We prove only one direction of the theorem, that if A is nonsingular then $\det(A) \neq 0$ and $\det(A^{-1}) = 1/\det(A)$. Suppose that A is non-singular. Then $AA^{-1} = I$. Taking determinants of both sides, and using that I is diagonal, so that $\det(I) = 1$, we see that

$$1 = \det(I) = \det(AA^{-1}) = \det(A)\det(A^{-1}).$$

Thus $\det(A) \neq 0$ (since otherwise the right hand side of this equation would be zero), and rearranging gives $\det(A^{-1}) = 1/\det(A)$.

Theorem

If A is a square matrix, then $\det(A) = \det(A^T)$.

Definition: Orthogonal Matrices

A square matrix A is orthogonal if $A^T = A^{-1}$.

Example: Determinants of orthogonal matrices (U2-LO7)

If A is an orthogonal matrix, what are the possible values of $\det(A)$?

If A is orthogonal, then applying the determinant to both sides of the equation $I = AA^T$ gives

$$\begin{aligned} 1 &= \det(I) = \det(AA^T) \\ &= \det(A)\det(A^T) = \det(A)^2. \end{aligned}$$

Taking square roots gives that $\det(A) = \pm 1$.